

$$Q_1 = A\bar{B}CD + \bar{A}\bar{C}\bar{D} + \bar{B}C + A\bar{C}D + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}D$$

$$= A\bar{B}CD + \bar{A}\bar{C}\bar{D}B + \bar{A}\bar{C}\bar{D}\bar{B} + ABC + \bar{A}\bar{B}C + A\bar{B}\bar{C}D + A\bar{B}\bar{C}D$$

$$+ \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}D$$

$$= A\bar{B}CD + \bar{A}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}\bar{B}\bar{C}D$$

$$+ A\bar{B}\bar{C}D + A\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}D$$

$$= \cancel{A\bar{B}CD} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D} + \bar{A}\bar{B}CD +$$

$$\cancel{1011} \quad 0100 \quad 0000 \quad 1011 \quad 1010 \quad 0011$$

$$\bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}D + A\bar{B}\bar{C}D + \cancel{\bar{A}\bar{B}C\bar{D}} = \sum m(0, 2, 3, 4, 9, 11, 10, 13)$$

$$0010 \quad 1101 \quad 1001 \quad \cancel{0010}$$

		CD			
		00	01	11	10
AB	00	1	0	1	1
	01	1	0	0	0
	11	0	1	0	0
	10	0	1	1	1

$$SOP \rightarrow \bar{A}\bar{C}\bar{D} + \bar{B}C + A\bar{C}D$$

$$POS \rightarrow (A + C + \bar{D}) \cdot (\bar{B} + C) \cdot (\bar{A} + C + D)$$

Date: _____

POS:

C_1	C_2	C_3
0001	0111	1100
0101	0110	1110
$(A+C+\bar{D})$	$(\bar{B}+\bar{C})$	$(\bar{A}+\bar{B}+\bar{C}+D)$

$$= (A+C+\bar{D})(\bar{B}+\bar{C})(\bar{A}+\bar{B}+\bar{C}+D)$$

Q NO2

from circuit:

$$= (\bar{A}+B+C)(C+\bar{D})(\bar{B}+\bar{C}+D)$$

$$\begin{array}{c} 0 \text{---} 1 \\ 1 \text{---} 0 \end{array}$$

$$1 \ 0 \ 0 \quad 0 \ 1 \quad 1 \ 1 \ 0$$

$$= (\bar{A}+B+C+D.\bar{D})(\bar{A}.A+C+\bar{D})(\bar{A}.A+\bar{B}+\bar{C}+D)$$

$$= (\bar{A}+B+C+D)(\bar{A}+B+C+\bar{D})(\bar{A}+C+\bar{D})(A+C+\bar{D})(\bar{A}+\bar{B}+\bar{C}+D)(A+\bar{B}+\bar{C}+D)$$

$$= (\bar{A}+B+C+D)(\bar{A}+B+C+\bar{D})(\bar{A}+B+C+\bar{D})(\bar{A}+\bar{B}+C+\bar{D})(A+B+C+\bar{D})(A+\bar{B}+C+\bar{D})(\bar{A}+\bar{B}+\bar{C}+D)(A+\bar{B}+\bar{C}+D)$$

$$1000, 1001, 1001, 1101, 0001, 0101, 1110, 0110$$

AB \ CD	00	01	11	10
00	1	0	1	1
01	1	0	1	0
11	1	0	1	0
10	0	0	1	1

C_1	C_2	C_3
0001	1000	0110
0101	1001	1110
1101		
1001		

$$(C+\bar{D}).(\bar{A}+B+C).(\bar{B}+\bar{C}+D)$$



~~my work~~
Standardize POS:

$$X = (\bar{A} + B + C + D) \cdot (\bar{A} + B + C + \bar{D}) \cdot (\bar{A} + \bar{B} + C + \bar{D}) \cdot (A + B + C + \bar{D}) \cdot (A + \bar{B} + C + \bar{D}) \cdot (\bar{A} + \bar{B} + \bar{C} + \bar{D}) \cdot (A + \bar{B} + \bar{C} + D)$$

Minimal from K. MAP:

$$(\bar{A} + B + C) \cdot (B + \bar{C} + D) \cdot (C + \bar{D})$$

Q₃

$$X = A\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}C\bar{D}\bar{E} + ABCDE + \bar{A}\bar{B}\bar{C}\bar{D}E + \bar{A}\bar{B}C\bar{D}E + A\bar{B}\bar{C}D\bar{E} + \bar{A}B\bar{C}\bar{D}\bar{E} + AB\bar{C}\bar{D}E$$

$$X = \sum m(0, 4, 6, 17, 26, 27, 31)$$

Plane 1: A = 0

BC \ DE	00	01	11	10
00	1 ₀	0 ₁	0 ₃	0 ₂
01	1 ₄	0 ₅	0 ₇	1 ₆
11	0 ₁₂	0 ₁₃	0 ₁₅	0 ₁₄
10	1 ₈	0 ₉	0 ₁₁	0 ₁₀

Plane: A = 1

BC \ DE	00	01	11	10
00	0 ₁₆	1 ₁₇	0 ₁₉	0 ₁₈
01	0 ₂₀	0 ₂₁	0 ₂₃	0 ₂₂
11	0 ₂₈	0 ₂₉	1 ₃₁	0 ₃₀
10	0 ₂₄	0 ₂₅	1 ₂₇	1 ₂₆

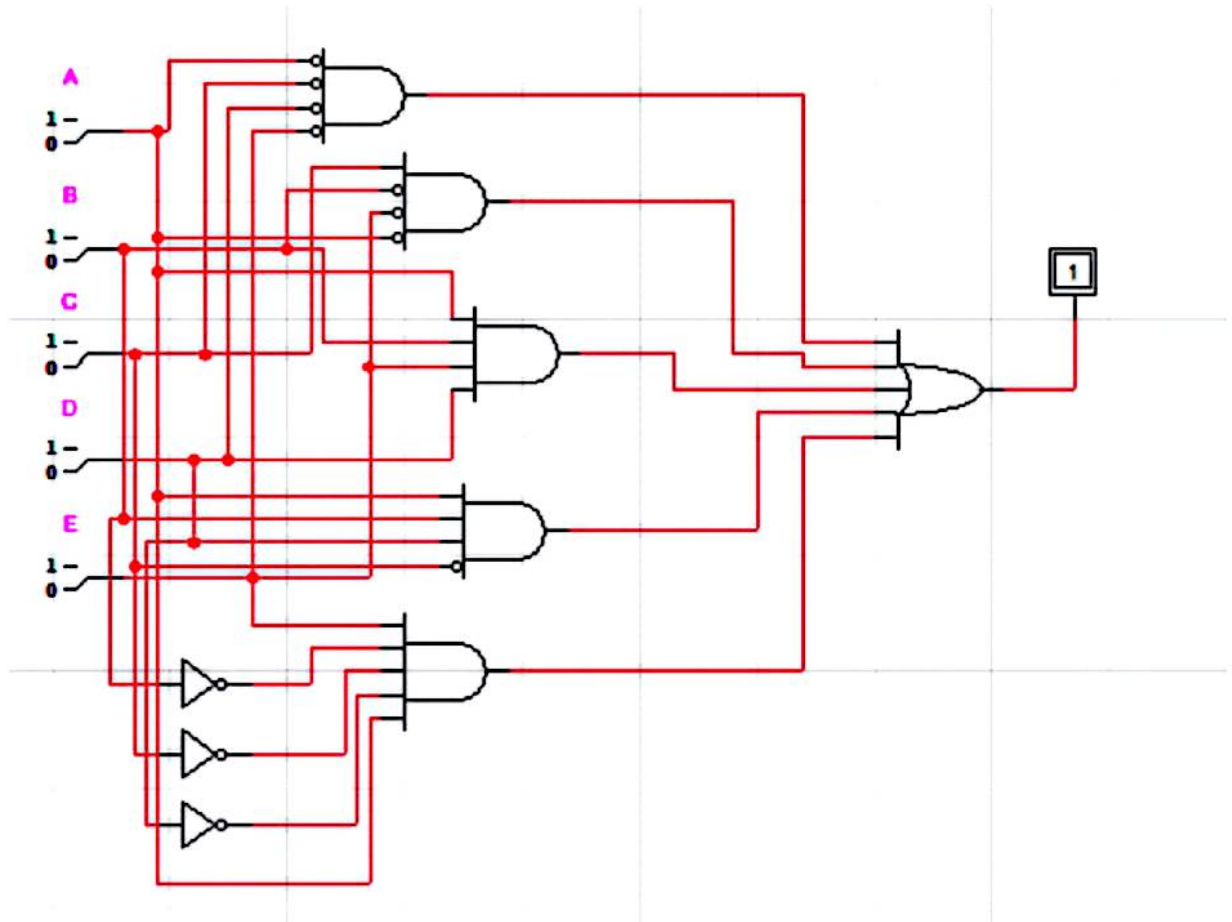
So, minimal SOP

G ₁	G ₂	G ₃	G ₄	G ₅
00000	00100	11011	11011	10001
01000	00110	11011	11011	

$$X = (\bar{A}\bar{B}\bar{C}\bar{D}\bar{E}) + (\bar{A}\bar{B}C\bar{D}\bar{E}) + (ABDE) + (A\bar{B}\bar{C}D) + (\bar{A}\bar{B}\bar{C}\bar{D}E)$$

A	B	C	D	E	Output (X)	Minterm
0	0	0	0	0	1	M0
0	0	1	0	0	1	M4
0	0	1	1	0	1	M6
0	1	0	0	0	1	M8
1	0	0	0	1	1	M17
1	1	0	1	0	1	M26
1	1	0	1	1	1	M27
1	1	1	1	1	1	M31

Final output is 1.



TT for standardize SOP:

Decimal (m)	A	B	C	D	E	Output (X)
0	0	0	0	0	0	1
4	0	0	1	0	0	1
6	0	0	1	1	0	1
8	0	1	0	0	0	1
17	1	0	0	0	1	1
26	1	1	0	1	0	1
27	1	1	0	1	1	1
31	1	1	1	1	1	1

Final output is 1.

Circuit

